AI-POWERED CROWD PREDICTION & SIMULATION

Intro

CrowdShield AI
Predict. Simulate. Protect.
AI-powered crowd prediction and decision-support simulator

Predict

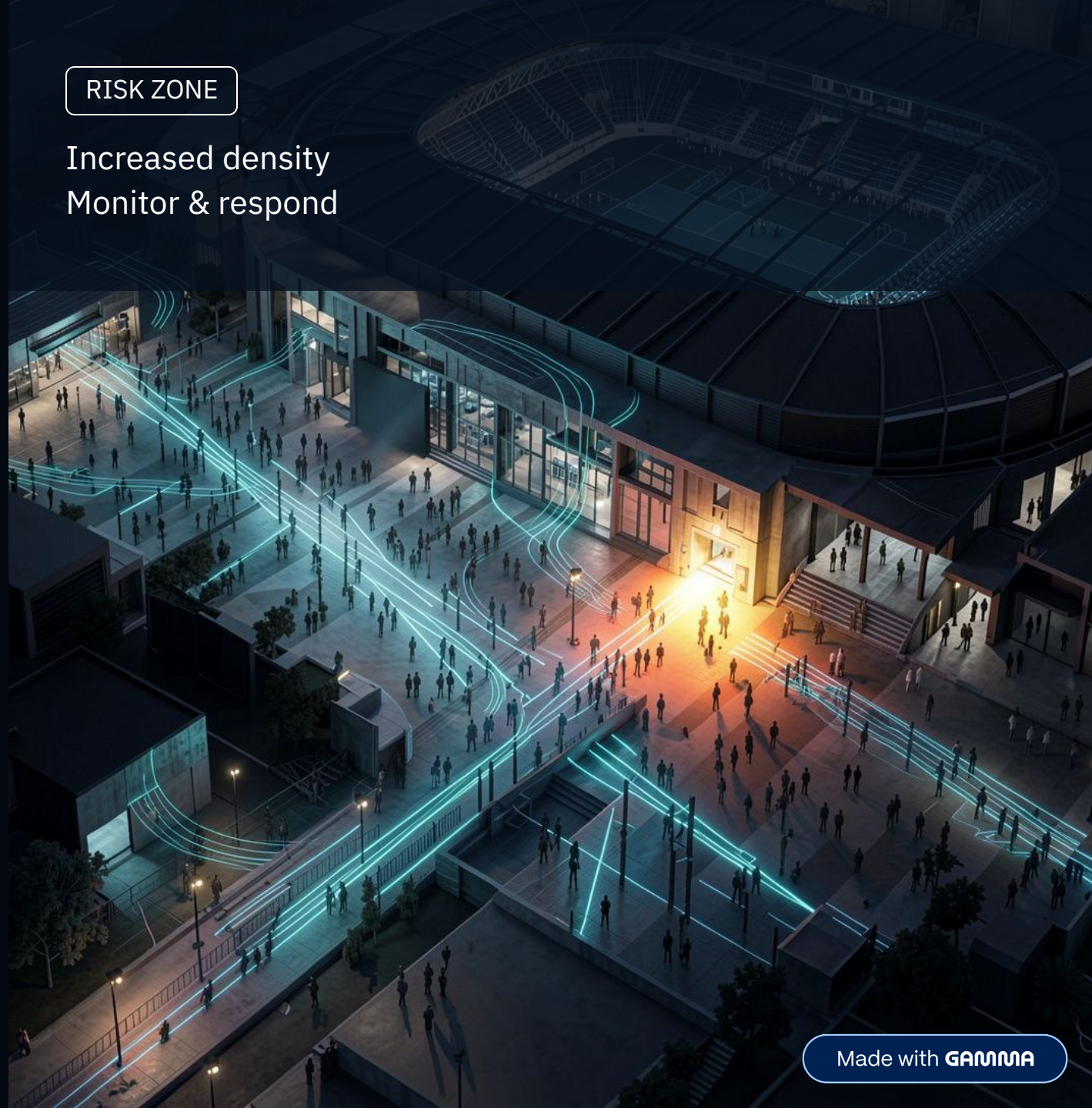
Where crowds will move

Simulate

How scenarios unfold

Decide

With confidence and speed



Made with GAMMA

THE CHALLENGE

Problem Statement

Crowds change continuously: normal flow can become dense, congested, and unstable as movement meets a bottleneck.

01



CALM FLOW

Normal movement and balanced distribution.

02



DENSITY BUILDUP

Rising density in a key area.

03



BOTTLENECK

Flow restriction creates pressure.

04



RISK

Potential loss of control and safety incidents.

Most systems describe the current state through density or congestion detection.

Operators need visibility into plausible next states and the consequences of acting—or waiting.

Made with GAMMA

THE APPROACH

Proposed Solution

01

Crowd video

Capture the live or uploaded scene.

02

AI analysis

Understand people, movement, and density.

03

Prediction

Estimate near-term crowd states.

04

Digital Twin

Reconstruct the environment in 3D.

05

What-if simulation

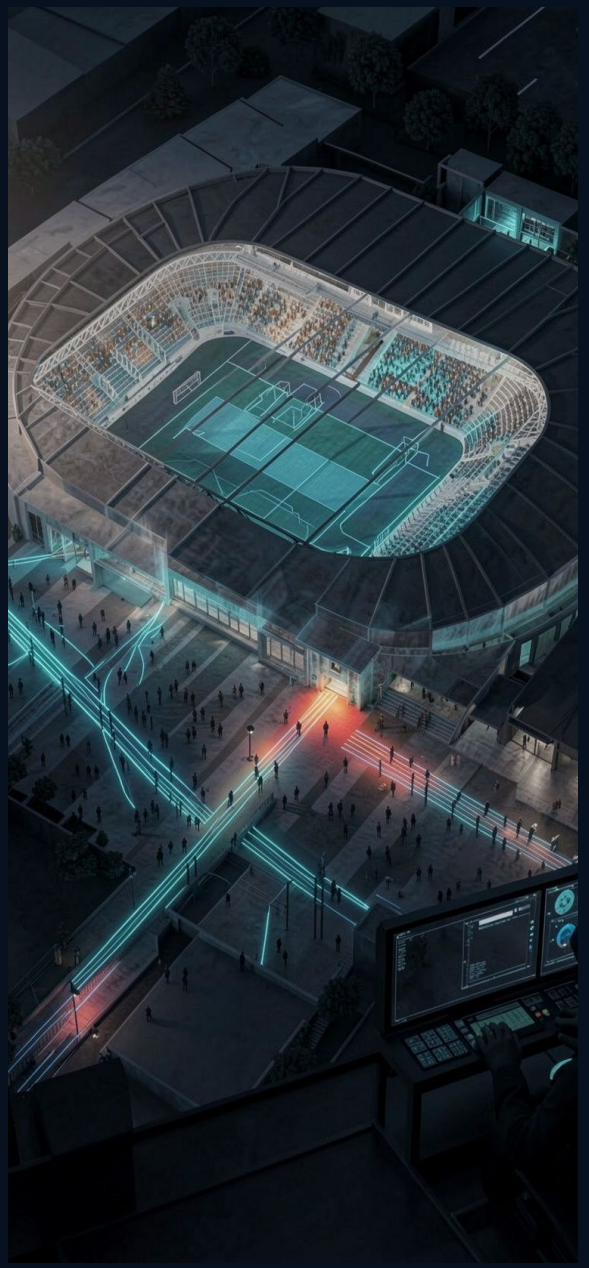
Test interventions before acting.

06

Human decision

Compare options and decide.

From seeing the crowd now to evaluating possible futures before acting.



The Digital Twin reveals possible futures.

Made with GAMMA

SUCCESS CRITERIA

Project Objectives



Detect

Detect people and estimate the current crowd state from video.



Predict

Predict near-term movement, density, and congestion risk.



Reconstruct

Reconstruct the venue as a 3D Digital Twin.



Simulate

Simulate alternative interventions without changing the real environment.



Compare

Compare scenarios using risk, flow, and evacuation indicators.

RISK-AWARE



Support

Give operators evidence-based recommendations and uncertainty indicators.

Made with GAMMA

DATA FOUNDATION

Database Collection & Description

01

Video sources

Publicly available crowd-video datasets; properly authorized or simulated event footage for prototype development.

02

Sampled frames

Frame sampling, resizing, annotation, normalization, and tracking-ready formatting.

03

Detections

People count,density and zone occupancy.

04

Scene context

Entrances, exits, corridors, bottlenecks, barriers, and control points.

05

Event labels

Congestion, buildup, flow change, and intervention scenario where available.

06

Model inputs

Live and scenario data for prediction and simulation.

Handling principle: privacy-preserving and authorized data practices throughout collection, preparation, and

Made with GAMMA

Tech Stack

Experience layer

Dashboard: React or Streamlit

Digital Twin: Three.js/WebGL or Unity

Application layer

Backend: Python and FastAPI

Data: SQLite or PostgreSQL

Intelligence layer

Vision: OpenCV, a YOLO-class detector, and multi-object tracking

Prediction: PyTorch or TensorFlow time-series/spatiotemporal model

Simulation & deployment

Simulation: Python-based or agent-based crowd simulation

Runtime: Docker; local GPU workstation or cloud

Prototype-friendly options; final choices depend on scope and available hardware.

Made with GAMMA

SYSTEM OVERVIEW

Architecture

A unified platform ingests live data, predicts crowd behavior, and makes defensible actions visible to operators.

DATA INPUTS

CCTV / video · Event data · External context

AI ENGINE

Data Fusion & Processing

Prediction model → simulation engine

DECISION SUPPORT

Risk Analysis & Scenario Engine

Intervention parameters feed the scenario engine.

OUTPUTS

Dashboard & Alerts

Density map · predicted flow · hotspots · risk score

Historical and scenario data inform prediction and risk analysis.

Operators see the environment, the forecast, and the consequence of each option.



Made with GAMMA

Proposed Methodology

01

Ingest

Crowd video.

02

Track

Detect and track people; estimate density and flow features.

03

Estimate

Current crowd state.

04

Predict

Short-horizon movement and congestion risk.

05

Model

3D Digital Twin; inject current state.

06

Compare

What-if interventions; compare outcomes; risk-aware decision support.

Current state + environment + intervention

→ simulated future state → risk comparison

PROBABILISTIC

Outputs: predicted density bottleneck probability, evacuation estimate, and uncertainty band.

Predictions are probabilistic decision-support outputs—not guaranteed outcomes.

Made with GAMMA

BUILD ENVIRONMENT

Software Requirements

Prototype guidance only; requirements vary with model, video resolution, and workload.

Software

- Python 3.x, FastAPI, OpenCV
- PyTorch or TensorFlow
- YOLO-class detector and tracking library/algorithm
- NumPy/Pandas and Three.js/WebGL or Unity
- PostgreSQL or SQLite, Git/GitHub, and Docker

Hardware

- Modern CPU laptop or workstation
- NVIDIA GPU recommended for training and inference acceleration
- 16 GB RAM recommended for prototype-scale workflows
- Sufficient storage for video datasets

Made with GAMMA

THE RESULT

Output Deliverables

Crowd-video analysis

Live or uploaded.

Density & flow

Crowd-density visualization.

Risk prediction

Short-term crowd risk; uncertainty indicators.

3D Digital Twin

For a venue or environment.

What-if simulator

Compare interventions.

Operator dashboard

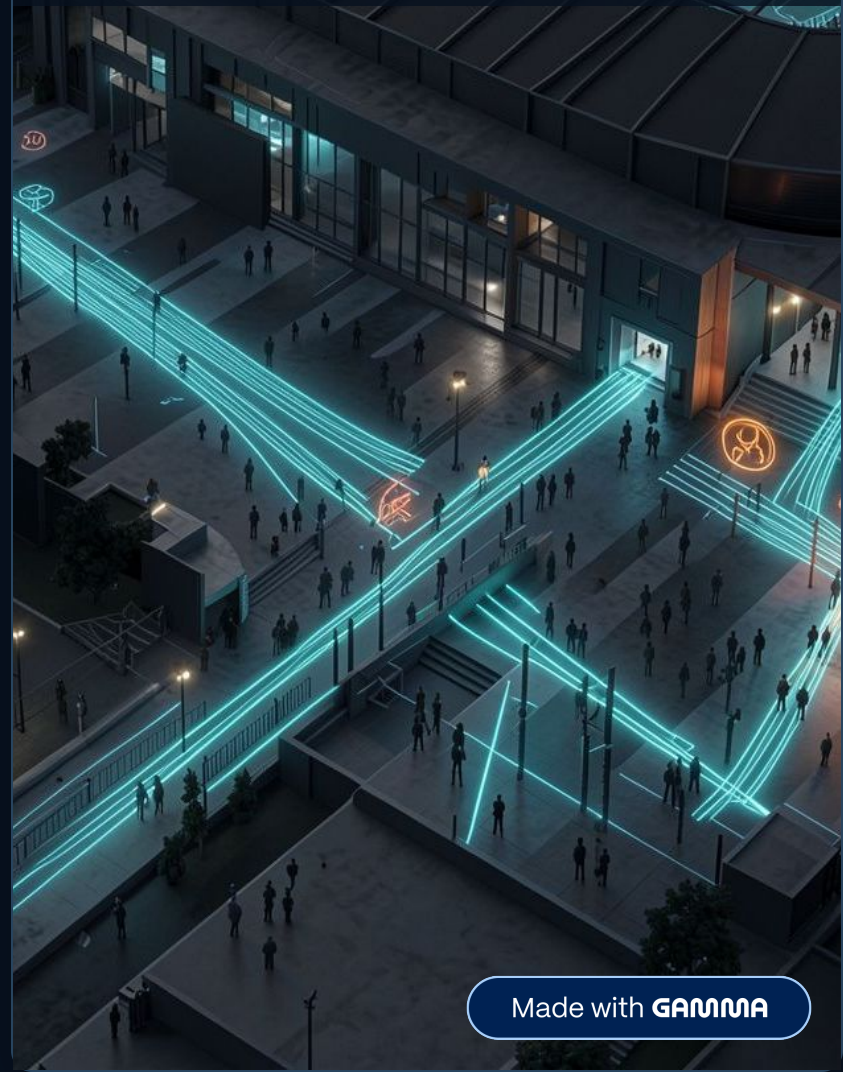
Risk indicators and recommendation support.

OPERATORS REMAIN IN CONTROL

Recommendations support informed human decisions.

Don’t Just See the Crowd. See What Happens Next.

CrowdShield AI — Predict. Simulate. Protect.



Made with GAMMA